

Title: Ethics and Data Science, Patil, Data Question

Assignment 4

by

Isaac Adebayo

MSDS 575: Ethics in Data Science

Dr Asmah Mullaem and Dr Gary Todd

Submission Date: February 23, 2026



DEPARTMENT OF COMPUTER SCIENCE AND DATA SCIENCE
SCHOOL OF APPLIED COMPUTATIONAL SCIENCES MEHARRY
MEDICAL COLLEGE 1005 DR. DB TODD JR BLVD, NASHVILLE,
TN 37208

1. What are the security challenges with data privacy?

Data scientist and software engineers' developers' intention is not to harm the people using their products. Code of data ethics instruct you not to collect data from subjects without informed consent but doesn't tell you how to implement the informed consent.

Challenge of data privacy

- Classes taught on ethics are not directly applicable to the real world
- Technology often outpaces the ethics model currently

2. What are the guiding principles of data-driven applications?

The five C

- **Consent**
Agreement starts with obtaining consent. Agreement between service provider and users are binary making it non-negotiable (meaning if you do not agree to the consent that may state your data might be used in many ways or sold, you will be denied service from using those providers)
- **Clarity**
Users should be told clearly what they are consenting to, what they are providing, what is going to be done with the data and the downstream consequence of how their data is used.
- **Consistency and Trust**
Trust requires consistency over time. Some people might have the best intention, but they may not honor this intention when you need them to or may misinterpret their intention. Once broken, rebuilding the trust may take a long time.

Consistency and trust may be broken explicitly or implicitly
- **Control and Transparency**
GDPR rule was enacted in Europe to give users more transparency and control over their data allowing them to change their mind on how their data is being used.
- **Consequence**
GINA (Genetic Information Non-discrimination Act) was put in law in 2008 to prevent genetic test screening from being used against an individual or their family

3. What is the role of technology and security in data?

Technology

- Evolution of technology affect data security

Security

- Why? Poor security led to vulnerability of data
- How? Human error
- Prevention? Checklist Manifesto to avoid common security compromise leading to data vulnerability
- Steps? Security training incorporated in school curriculum rather than a separate course (e.g. SQL, SQL injection mitigation)
 - bug bounty program to assess code and data vulnerability
 - Ethical hiring
 - Able to escalate violation without the fear of retaliation, reprisal or remediation

4. What does the future hold for data, security, and privacy?

Regulation and fine

- GDPR model for data privacy and consent,
- Nuremberg code based on NAZI atrocities regarding to consent on human experiment

Conversation

- Awareness about data usage
- Autonomy regarding ethical usage of their data
- Consent: Explicit use of their personal data

5. What are some security challenges with data and chatbots?

Challenges

- Chatbot: Balancing or trade off between privacy, liberty, entrapment, accountability and compliance with international law
- Data: Privacy and handling of data particularly data with individual who cannot make well informed consent such as minors in the school system that amass lot of personal data